

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; kimi-k2.7-code

Abstract

This report evaluates `moonshotai/kimi-k2.7-code` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron leads Cache hit rate in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie; Infron leads Observed cost in all routing modes; Infron leads Throughput in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; OpenRouter leads E2E latency in all routing modes; OpenRouter leads Streaming TTFT in all routing modes.

Overall, Infron's cross-mode strengths are observed cost, cache reuse, while OpenRouter's cross-mode strengths are throughput, Streaming TTFT, E2E latency. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform–level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput OpenRouter wins 3/4 Max advantage 122.01%, higher is better	Observed cost Infron wins 4/4 Max advantage 16.92%, lower is better	E2E latency OpenRouter wins 4/4 Max advantage 1.81%, lower is better	Streaming TTFT OpenRouter wins 4/4 Max advantage 1.64%, lower is better	Cache hit rate Infron wins 3/4 Max advantage 6.75%, higher is better
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Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	Infron advantage 35.79%	Infron advantage 2.04%	OpenRouter advantage 63.38%	OpenRouter advantage 51.25%	OpenRouter advantage 6.75%
Price First price	OpenRouter advantage 122.01%	Infron advantage 16.92%	OpenRouter advantage 1.81%	OpenRouter advantage 1.64%	Infron advantage 1.57%
Latency First latency	OpenRouter advantage 65.03%	Infron advantage 12.86%	OpenRouter advantage 75.27%	OpenRouter advantage 27.96%	Infron advantage 1.44%
TTFT First ttft	OpenRouter advantage 92.77%	Infron advantage 3.38%	OpenRouter advantage 102.32%	OpenRouter advantage 37.09%	Infron advantage 0.36%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second <code>usage.prompt_tokens</code> deltas are limited to 50 tokens within each <code>sort/group/round</code> pair.	Methods and data quality
Cache reuse	Infron leads Cache hit rate in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in all routing modes	Overall metrics and provider drill-down
Performance	Infron leads Throughput in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; OpenRouter leads E2E latency in all routing modes; OpenRouter leads Streaming TTFT in all routing modes	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	OpenRouter	Infron	Infron	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	Infron	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Latency First	Minimize full–response waiting time	Infron	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
TTFT First	Minimize streaming first–token time	Infron	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference–platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first–token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token–level cache hit rate for repeated stable prefixes.	Second–request cache–read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response–returned `usage.prompt_tokens` as the input–token control while allowing small cross–platform accounting variance up to 50 tokens.
- Extends prompt–caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

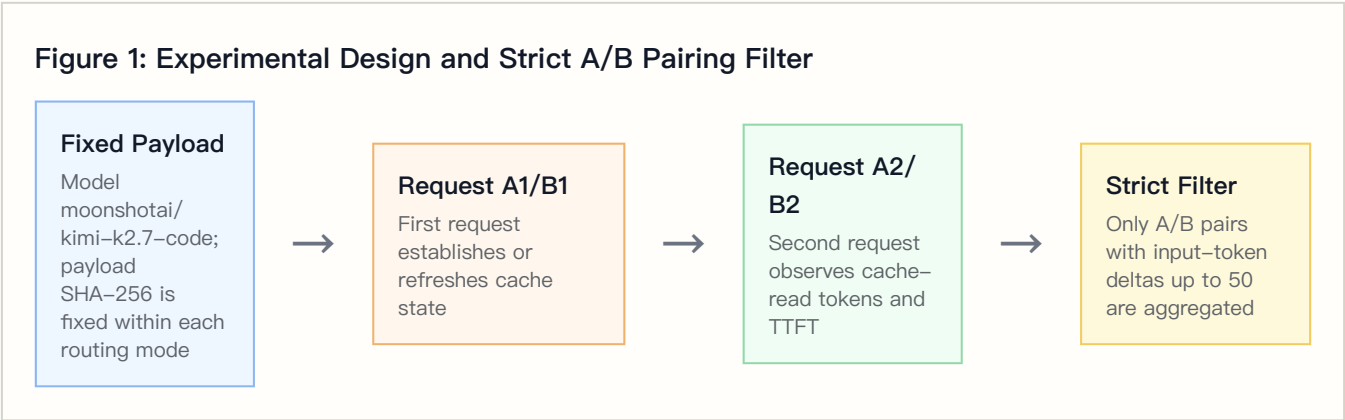
2. Experimental Design, Dataset, and Controls

2.1 Dataset Construction

The dataset is `controlled_cache_probe`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with <code>cache_read_tokens > 0</code>	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	moonshotai/kimi-k2.7-code
Provider model IDs	infron: moonshotai/kimi-k2.7-code ; openrouter: moonshotai/kimi-k2.7-code
Platforms	Infron and OpenRouter
API protocol	/v1/chat/completions
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	short ≈1500, medium ≈8000, long ≈32000
Excluded records	0

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	200	6260516	91.63%	\$1.19081600	6.40 tok/s	4905.39 ms	3968.00 ms
Throughput First	OpenRouter	200	6260516	97.81%	\$1.21514801	4.71 tok/s	3002.51 ms	2623.43 ms
Price First	Infron	200	6260516	99.83%	\$1.01834100	1.98 tok/s	9519.19 ms	7739.13 ms
Price First	OpenRouter	200	6260516	98.29%	\$1.19068547	4.40 tok/s	3388.53 ms	2931.51 ms
Latency First	Infron	200	6260516	99.93%	\$0.99795500	2.83 tok/s	5646.34 ms	3509.06 ms
Latency First	OpenRouter	200	6260516	98.51%	\$1.12628891	4.68 tok/s	3221.60 ms	2742.34 ms
TTFT First	Infron	200	6260516	99.87%	\$1.07852900	2.46 tok/s	6499.95 ms	3668.66 ms
TTFT First	OpenRouter	200	6260516	99.52%	\$1.11493858	4.75 tok/s	3212.64 ms	2676.15 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	4075.03 ms	9776.66 ms	17941.59 ms	3081.10 ms	8616.90 ms	16370.53 ms
Throughput First	OpenRouter	2501.84 ms	5158.99 ms	15671.93 ms	2116.70 ms	4460.52 ms	15601.83 ms
Price First	Infron	5348.32 ms	45655.32 ms	50382.55 ms	3366.74 ms	43087.14 ms	49519.27 ms
Price First	OpenRouter	2804.84 ms	6482.46 ms	14736.20 ms	2368.34 ms	5565.66 ms	14332.86 ms
Latency First	Infron	5460.02 ms	8440.44 ms	10015.20 ms	3096.11 ms	6159.66 ms	7641.69 ms
Latency First	OpenRouter	2851.02 ms	5259.72 ms	8360.13 ms	2405.65 ms	4679.84 ms	7342.31 ms
TTFT First	Infron	5230.28 ms	23478.28 ms	32480.48 ms	2992.13 ms	7981.20 ms	16022.86 ms
TTFT First	OpenRouter	2957.06 ms	5520.04 ms	7872.32 ms	2370.26 ms	4557.33 ms	6996.73 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	–3805.76 ms	–4741.72 ms to –2946.64 ms	<0.001	200	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	–2689.12 ms	–3629.63 ms to –1853.28 ms	<0.001	200	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	1.4418 tok/s	1.0028 tok/s to 1.8684 tok/s	<0.001	200	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	\$0.00012166	\$–0.00104370 to \$0.00120686	0.8403	200	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	–4.50 pp	–8.15 pp to –0.90 pp	0.0155	200	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	–12261.33 ms	–14936.97 ms to –9598.65 ms	<0.001	200	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	–9615.23 ms	–12319.47 ms to –6803.63 ms	<0.001	200	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	–2.4544 tok/s	–2.7099 tok/s to –2.1819 tok/s	<0.001	200	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	\$0.00086172	\$0.00033375 to \$0.00147840	<0.001	200	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	1.28 pp	–0.25 pp to 3.07 pp	0.2184	200	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	–4849.49 ms	–5417.67 ms to –4220.62 ms	<0.001	200	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	–1533.43 ms	–2057.26 ms to –912.79 ms	<0.001	200	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	-2.4749 tok/s	-2.7115 tok/s to -2.2312 tok/s	<0.001	200	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	\$0.00064167	\$0.00042288 to \$0.00098094	<0.001	200	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	2.11 pp	1.00 pp to 3.67 pp	<0.001	200	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	-6574.62 ms	-7946.94 ms to -5311.26 ms	<0.001	200	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	-1985.02 ms	-2658.57 ms to -1336.57 ms	<0.001	200	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	-2.0202 tok/s	-2.3307 tok/s to -1.7058 tok/s	<0.001	200	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	\$0.00018205	\$0.00005417 to \$0.00031947	0.0072	200	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	2.71 pp	1.10 pp to 4.79 pp	<0.001	200	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	11204	28.0100	400	4112.96 ms	3968.00 ms	4905.39 ms
Throughput First	OpenRouter	5829	14.5725	400	2623.43 ms	2623.43 ms	3002.51 ms
Price First	Infron	2084	5.2100	75	7939.56 ms	7739.13 ms	9519.19 ms
Price First	OpenRouter	6254	15.6350	400	2931.51 ms	2931.51 ms	3388.53 ms
Latency First	Infron	0	0.0000	0	3716.89 ms	3509.06 ms	5646.34 ms
Latency First	OpenRouter	6300	15.7500	400	2742.34 ms	2742.34 ms	3221.60 ms
TTFT First	Infron	2144	5.3600	134	3980.68 ms	3668.66 ms	6499.95 ms
TTFT First	OpenRouter	6357	15.8925	400	2676.15 ms	2676.15 ms	3212.64 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions` ; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
chat_completions	/v1/chat/completions	Infron	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%
chat_completions	/v1/chat/completions	OpenRouter	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

**Figure 3–E2:
Normalized
capability
contour**

Five metrics are normalized to 0–100, where higher is better.

**Figure 3–E3:
Cost–cache
efficiency plane**

X-axis is token cache hit rate; Y-axis is observed total cost.

**Figure 3–E1:
Routing–
mode
core
metric
comparison**

Infron and OpenRouter are shown side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache–domain and performance differences.

Routing–Mode Drill–Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

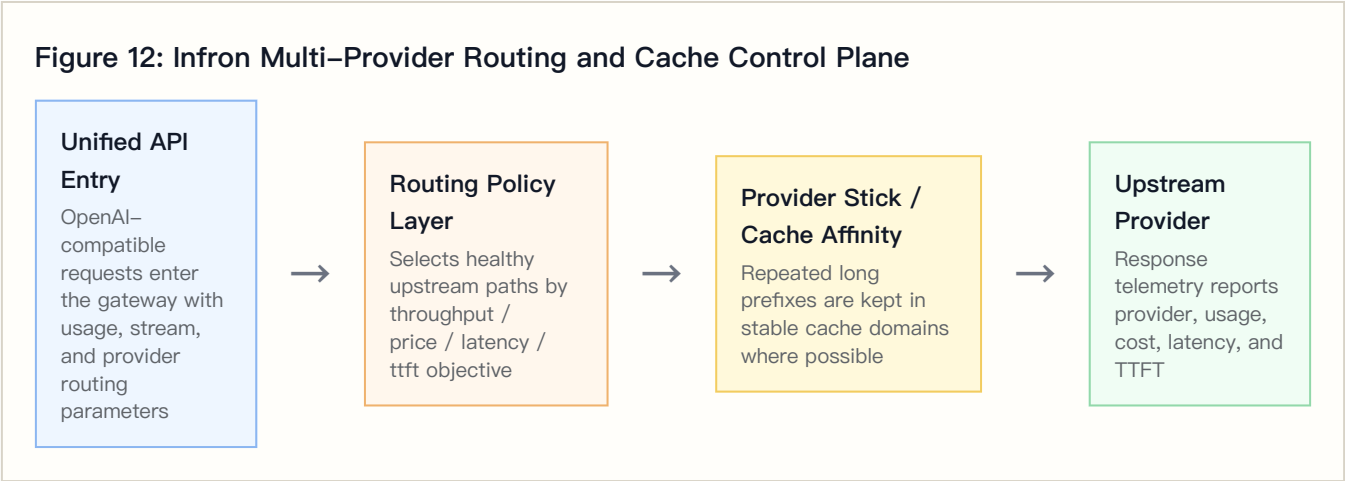
Figure 4–E3: Latency First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	400	400	alibaba/eu 144, together 129, alibaba/us 64, alibaba/sg 57, alibaba/cn 6

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	OpenRouter	400	400	ModelRun 252, Ambient 104, Together 44
Price First	Infron	400	400	deepinfra 319, alibaba/cn 52, parasail 23, inception 6
Price First	OpenRouter	400	400	ModelRun 152, Ambient 130, Together 50, DeepInfra 40, Inception 21, Parasail 5, Venice 2
Latency First	Infron	400	400	deepinfra 400
Latency First	OpenRouter	400	400	Ambient 130, ModelRun 128, Together 53, SiliconFlow 35, DeepInfra 28, Inception 26
TTFT First	Infron	400	400	deepinfra 266, together 134
TTFT First	OpenRouter	400	400	Ambient 128, ModelRun 102, SiliconFlow 81, Together 54, Inception 22, DeepInfra 13

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/ second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Throughput First	Infron	alibaba/eu	144	36.00%	78/66	79	3286.44 ms	4366.93 ms	2110373	5624
Throughput First	Infron	together	129	32.25%	57/72	72	3235.42 ms	4006.65 ms	2066908	2064
Throughput First	Infron	alibaba/us	64	16.00%	31/33	51	5265.20 ms	6283.41 ms	1048741	2453
Throughput First	Infron	alibaba/sg	57	14.25%	31/26	44	4472.96 ms	5327.93 ms	939968	2200
Throughput First	Infron	alibaba/cn	6	1.50%	3/3	3	17441.67 ms	18438.28 ms	94526	218
Throughput First	OpenRouter	ModelRun	252	63.00%	144/108	149	2519.72 ms	2829.30 ms	3886150	3294
Throughput First	OpenRouter	Ambient	104	26.00%	37/67	72	2841.15 ms	3429.31 ms	1907117	1664
Throughput First	OpenRouter	Together	44	11.00%	19/25	25	2702.84 ms	2985.72 ms	467249	704
Price First	Infron	deepinfra	319	79.75%	153/166	167	3350.54 ms	5166.63 ms	5108067	5104
Price First	Infron	alibaba/cn	52	13.00%	24/28	41	20123.07 ms	21256.93 ms	778008	1976

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Price First	Infron	parasail	23	5.75%	17/6	23	42034.05 ms	45037.46 ms	341741	368
Price First	Infron	inception	6	1.50%	6/0	6	2274.74 ms	3050.34 ms	32700	96
Price First	OpenRouter	ModelRun	152	38.00%	86/66	86	2323.12 ms	2596.69 ms	2255509	1993
Price First	OpenRouter	Ambient	130	32.50%	65/65	65	3160.64 ms	3698.92 ms	2392546	2080
Price First	OpenRouter	Together	50	12.50%	25/25	25	2744.26 ms	2955.57 ms	541920	800
Price First	OpenRouter	DeepInfra	40	10.00%	13/27	28	3006.17 ms	4022.11 ms	596328	640
Price First	OpenRouter	Inception	21	5.25%	8/13	13	5186.99 ms	5777.81 ms	432377	336
Price First	OpenRouter	Parasail	5	1.25%	2/3	5	4209.77 ms	5637.74 ms	30936	80
Price First	OpenRouter	Venice	2	0.50%	1/1	2	10586.56 ms	10835.09 ms	10900	32
Latency First	Infron	deepinfra	400	100.00%	200/200	200	3509.06 ms	5646.34 ms	6260516	6400
Latency First	OpenRouter	Ambient	130	32.50%	65/65	65	3266.40 ms	3766.15 ms	2392546	2080
Latency First	OpenRouter	ModelRun	128	32.00%	68/60	74	2150.04 ms	2469.19 ms	1684555	1675
Latency First	OpenRouter	Together	53	13.25%	26/27	27	2592.31 ms	3085.15 ms	596555	848
Latency First	OpenRouter	SiliconFlow	35	8.75%	12/23	25	3137.07 ms	3641.41 ms	661817	560
Latency First	OpenRouter	DeepInfra	28	7.00%	16/12	16	2534.16 ms	3431.50 ms	380049	448
Latency First	OpenRouter	Inception	26	6.50%	13/13	13	3036.65 ms	3689.96 ms	544994	416
TTFT First	Infron	deepinfra	266	66.50%	139/127	139	3895.28 ms	7869.11 ms	4039020	4256
TTFT First	Infron	together	134	33.50%	61/73	73	3218.80 ms	3782.07 ms	2221496	2144
TTFT First	OpenRouter	Ambient	128	32.00%	64/64	64	2670.13 ms	3250.29 ms	2319820	2048
TTFT First	OpenRouter	ModelRun	102	25.50%	55/47	57	2354.88 ms	2652.64 ms	1386865	1331

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
TTFT First	OpenRouter	SiliconFlow	81	20.25%	36/45	46	2927.23 ms	3581.49 ms	1212014	1296
TTFT First	OpenRouter	Together	54	13.50%	27/27	27	2680.42 ms	3056.62 ms	632918	864
TTFT First	OpenRouter	Inceptron	22	5.50%	11/11	11	2971.92 ms	3536.26 ms	468740	352
TTFT First	OpenRouter	DeepInfra	13	3.25%	7/6	7	3173.49 ms	5038.10 ms	240159	208

7.1 Cache–Rate and Cost Divergence Drill–Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing–mode differences.

Routing mode	Cache–hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	–6.18 pp	0.98x	alibaba/eu 36.00%	ModelRun 63.00%	+5375	Cache and cost move in different directions; evaluate with speed metrics
Price First	+1.54 pp	0.86x	deepinfra 79.75%	ModelRun 38.00%	–4170	Infron leads both cache and cost
Latency First	+1.42 pp	0.89x	deepinfra 100.00%	Ambient 32.50%	–6300	Infron leads both cache and cost
TTFT First	+0.36 pp	0.97x	deepinfra 66.50%	Ambient 32.00%	–4213	Infron leads both cache and cost

8. Stratified Results: Prompt–Length Cache Performance

This section aggregates second–request cache–read tokens, token–level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt–length tier. Bold cells mark the advantaged side within each tier.

Prompt–Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	Infron	268	472752	458064	96.89%	\$0.18841300	5439.10 ms	3342.51 ms
short	1500	OpenRouter	268	472752	448288	94.83%	\$0.23035416	2322.02 ms	1856.68 ms

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
medium	8000	Infron	268	2448448	2413696	98.58%	\$0.83923300	6244.76 ms	4307.83 ms
medium	8000	OpenRouter	268	2448448	2409056	98.39%	\$0.92854540	3053.28 ms	2608.15 ms
long	32000	Infron	264	9599832	9375808	97.67%	\$3.25799500	8268.56 ms	6540.45 ms
long	32000	OpenRouter	264	9599832	9479792	98.75%	\$3.48816141	4259.38 ms	3780.74 ms

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
short	Throughput First	91.24%	95.85%	OpenRouter
short	Price First	97.38%	96.79%	Infron
short	Latency First	99.77%	95.33%	Infron
short	TTFT First	99.18%	91.33%	Infron
medium	Throughput First	95.02%	96.51%	OpenRouter
medium	Price First	99.78%	98.06%	Infron
medium	Latency First	99.82%	99.51%	Infron
medium	TTFT First	99.70%	99.50%	Infron
long	Throughput First	90.78%	98.24%	OpenRouter
long	Price First	99.96%	98.42%	Infron
long	Latency First	99.97%	98.42%	Infron
long	TTFT First	99.95%	99.92%	Infron

9. Stratified Results: Group–Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	99.42%	\$0.18490200	9118.37 ms	8160.20 ms
Infron	2	50	50	99.53%	\$0.16532800	11440.68 ms	9995.32 ms
Infron	3	50	50	68.31%	\$0.49768300	11416.34 ms	10017.37 ms
Infron	4	50	50	99.72%	\$0.34290300	6672.99 ms	5438.48 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
OpenRouter	1	50	50	93.24%	\$0.39442468	5391.73 ms	5149.60 ms
OpenRouter	2	50	50	99.69%	\$0.27490953	4662.05 ms	3847.97 ms
OpenRouter	3	50	50	98.55%	\$0.27949530	4042.35 ms	3624.04 ms
OpenRouter	4	50	50	99.63%	\$0.26631850	5440.47 ms	4903.39 ms

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	99.50%	\$0.26391400	50124.13 ms	49185.34 ms
Infron	2	50	50	99.93%	\$0.25545300	7555.98 ms	5646.10 ms
Infron	3	50	50	99.94%	\$0.25371500	7740.16 ms	5524.46 ms
Infron	4	50	50	99.93%	\$0.24525900	6585.08 ms	5379.35 ms
OpenRouter	1	50	50	94.99%	\$0.36678692	7138.74 ms	6752.04 ms
OpenRouter	2	50	50	99.79%	\$0.27203762	5370.75 ms	4930.24 ms
OpenRouter	3	50	50	99.71%	\$0.28004226	5607.14 ms	5391.60 ms
OpenRouter	4	50	50	98.55%	\$0.27181867	6567.82 ms	5522.28 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	99.93%	\$0.24495500	8821.03 ms	7302.37 ms
Infron	2	50	50	99.93%	\$0.25535600	8146.96 ms	5933.52 ms
Infron	3	50	50	99.94%	\$0.25305000	9371.44 ms	6168.72 ms
Infron	4	50	50	99.93%	\$0.24459400	6745.53 ms	5387.75 ms
OpenRouter	1	50	50	94.98%	\$0.29494020	6532.99 ms	5275.54 ms
OpenRouter	2	50	50	99.53%	\$0.28091353	4604.56 ms	3757.73 ms
OpenRouter	3	50	50	99.72%	\$0.28152824	5259.72 ms	4701.27 ms
OpenRouter	4	50	50	99.72%	\$0.26890694	5082.41 ms	3993.14 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	99.93%	\$0.24459400	31338.84 ms	12997.05 ms
Infron	2	50	50	99.93%	\$0.25535600	6212.88 ms	4393.05 ms
Infron	3	50	50	99.85%	\$0.27795100	6659.21 ms	5191.00 ms
Infron	4	50	50	99.78%	\$0.30062800	5462.57 ms	4847.95 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
OpenRouter	1	50	50	99.27%	\$0.27005351	5624.37 ms	4442.18 ms
OpenRouter	2	50	50	99.33%	\$0.28458315	4943.42 ms	4571.01 ms
OpenRouter	3	50	50	99.74%	\$0.28763987	5516.58 ms	4374.55 ms
OpenRouter	4	50	50	99.73%	\$0.27266205	5244.23 ms	4892.93 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	OpenRouter leads overall (3/5 metrics)	Batch generation, offline summaries, backend processing	First-token and E2E response are faster; check cache and cost
Price First	Minimize request and token cost	OpenRouter leads overall (3/5 metrics)	High-frequency templates, support automation, marketing, RAG prefixes	Cache and cost are stronger; still check speed SLA
Latency First	Minimize full-response waiting time	OpenRouter leads overall (3/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	Cache and cost are stronger; still check speed SLA
TTFT First	Minimize streaming first-token time	OpenRouter leads overall (3/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	Cache and cost are stronger; still check speed SLA

11. Conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	OpenRouter	Infron	Infron	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	Infron	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Latency First	Minimize full-response waiting time	Infron	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
TTFT First	Minimize streaming first-token time	Infron	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short-window stability but not day-level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Online link
Summary	summary.json
Paired dataset	benchmark_pairs.csv
Request-level dataset	benchmark_requests.jsonl
Filtered structured records	records.json
Excluded-record audit	records_excluded.json
Benchmark runner source	rerun_routing_sort_cache_cost_ab.py
HTML report renderer source	render_glm52_deepseek_style_report.py
Test source	test_rerun_routing_sort_cache_cost_ab.py
Dataset reference	data directory